BATCH 9 · COMPLETED BEFORE ANY CODE WAS WRITTEN

CBIM Problem Canvas

Credit Pre-Screening Engine — Loan Intake Optimisation · MUC02 · Path 2, KLU Batch

Completed together by all four members in a 20-minute session before the dataset was opened. The canvas is reproduced identically at the top of the notebook.

Situation (S)

A mid-size private bank's retail lending unit receives about **3,000 unsecured personal loan applications a month** from salaried, self-employed, business-owner and gig applicants across Tier 1–3 cities. Every application is reviewed manually by a credit analyst, taking roughly **25 minutes each** — about **1,250 analyst hours a month**.

Complication (C)

About **47% of applications are declined**, and a large share of those are decided in the first minute because the credit score is far below the bank's working floor or existing EMIs already consume most of the applicant's income. Three consequences follow: analyst time is spent on clear-cut declines rather than on borderline cases that need judgement; good applicants wait longer in the queue behind files nobody needed to read; and the bank cannot grow intake without hiring proportionally.

Question (Q)

Which applications can be removed from the manual review queue automatically, and at what model-confidence threshold, without wrongly declining more than 5% of applicants the credit team would have approved?

The decision this enables is operational, not analytical: the bank introduces an auto-decline lane at a stated confidence level and re-deploys the freed analyst hours onto applications that genuinely need a human.

SSOT — single source of truth

`MUC02_Credit_PreScreening_Dataset.csv` — 12,000 applications received between April and September 2026, one row per application, 19 columns, with the credit team's final decision recorded for each.

The critical qualification: the `decision` column is the credit team's past judgement, not whether the customer repaid. The model therefore learns to reproduce historical analyst behaviour, including any inconsistency or bias in it. Model error is measured against what an analyst would have decided, not against actual credit risk.

Success definition

Not accuracy. The pre-screener is good enough when, at the chosen threshold, **no more than 5% of auto-declined applications are ones the credit team would have approved** — the bank's stated risk appetite — while the volume removed is large enough to matter, stated explicitly in **applications per month and analyst hours per month**. A secondary condition: the model must not be trusted on segments where its accuracy is near chance, and those segments must be named.

Primary stakeholder

The **Credit Policy Analyst**, who owns the recommendation and must defend it to the Head of Retail Lending.

Also affected: the **Head of Retail Lending** (approves the change, wants hours and risk); the **credit analysts**, whose queue composition changes as easy declines leave and hard cases remain; and the **applicants**, in particular those declined without a human ever reading their file.